Three Dyes

Time limit: 2 seconds | Memory limit: 256 MB

The Weavers' Guild of Codeforcia makes long ribbons out of colored threads. A ribbon is described by a string s of lowercase Latin letters, where the i-th letter is the color of the i-th thread. The letter r stands for red, g for green, b for blue, and every other letter stands for some other color that the guild does not care about.

The guild considers a piece of ribbon **harmonious** if it contains the same number of red, green and blue threads. Note that this number is allowed to be zero, so for example a piece made only of yellow threads is harmonious.

Formally, a piece of the ribbon is a non-empty contiguous substring $s[l..r]$ with $1 \le l \le r \le |s|$. A piece is harmonious if the number of letters r, the number of letters g and the number of letters b in it are all equal.

Two pieces are considered different if they start or end at different positions, even if they are equal as strings. Your task is to count the harmonious pieces of the ribbon.

Since ribbons can become very long, the guild needs your help. Write a program to count the harmonious pieces for each ribbon.

Input

The first line of input contains a single integer T ($1 \le T \le 10^4$) representing the number of test cases. This is followed by a description of each test case.

Each test case consists of a single line containing a string s ($1 \le |s| \le 2 \cdot 10^5$) consisting of lowercase Latin letters.

It is guaranteed that the sum of $|s|$ across all the test cases will not exceed $2 \cdot 10^5$.

Output

For each test case, output a single integer: the number of harmonious pieces of the ribbon. Note that the answer can be large, so use a 64-bit integer type.

Examples

Input	Output
6 rgb	1 5

rgbrgb aaa rrggbb brgxrgb rrr	6 1 6 0

Note

In the first test case, the only harmonious piece is the whole ribbon `rgb`. A single letter such as `r` is not harmonious because it has one red thread and no green or blue threads.

In the second test case, the harmonious pieces are `rgb` (positions 1–3), `gbr` (2–4), `brg` (3–5), `rgb` (4–6) and the whole ribbon (1–6). The two pieces equal to `rgb` are counted separately.

In the third test case, there are no red, green or blue threads at all, so every one of the 6 pieces is harmonious.

In the fourth test case, the only harmonious piece is the whole ribbon `rrggbb`.

In the fifth test case, the harmonious pieces are `brg` (1–3), `brgx` (1–4), `x` (4–4), `xrgb` (4–7), `rgb` (5–7) and the whole ribbon (1–7).

In the sixth test case, no piece contains the same number of red, green and blue threads.